T3 Firmware & Bootloader

Skip firmware signature validation · built Nov 13, 2026

Skip firmware signature validation

Skip Firmware Signature Validation via Modbus Register 1992

This firmware supports bypassing firmware HMAC validation when the Modbus secure key register is written with the special value `0x0BEE`.

What happens

- Modbus register `1992` is mapped to `MODBUS_SECURE_KEY`.
- When the stored value equals `0x0BEE`, firmware validation is skipped during update.
- The firmware logs: `Skip firmware HMAC validation, secure key found in modbus register`.

Register details

- Register address: `1992`
- Expected bypass value: `0x0BEE`
- Data type: 16-bit register

How to write the bypass key

Use your Modbus master / RS485 tool to write register `1992` with value `0x0BEE`.

Example:

- Register: `1992`
- Value: `30574` decimal or `0x0BEE` hexadecimal
- Byte order in the register: the firmware reads the 16-bit value directly and stores it in flash

If your tool allows writing a register word, set it to `0x0BEE`. If your tool writes high/low bytes separately, send `0x0B` as the high byte and `0xEE` as the low byte.

How to verify

After writing the register:

1. Reboot or restart the update flow as needed.

2. Start firmware upload/update.
3. The device should skip HMAC validation and continue.

Important notes

- This bypass is intended for testing, recovery, or controlled validation scenarios only.
- Do not leave the bypass key enabled in production.
- For normal signed firmware validation, clear the register value or restore it to `0`.

Quick reference

```text
Register   : 1992
Bypass key : 0x0BEE
Result     : skip firmware HMAC validation
```